

Exponential semiaxes have an unavoidable square-root gap

A full counterexample to the question in arXiv:1901.06639

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11 August 2026

Status. Candidate full counterexample, likely valid, pending human review. The source's constant-factor comparison is false for every sequence of semiaxes comparable to a geometric progression.

1 Source question

Hinrichs, Krieg, Novak, Prochno, and Ullrich study the ellipsoid

$$\mathcal{E}_\sigma = \left\{ x \in \ell_2 : \sum_{j \geq 1} \left(\frac{x_j}{\sigma_j} \right)^2 \leq 1 \right\}$$

and the radius left after n independent standard Gaussian measurements,

$$\mathcal{R}_n(\sigma) = \sup\{\|x\|_2 : x \in \mathcal{E}_\sigma, G_n x = 0\}.$$

Optimal information has radius σ_{n+1} . On PDF page 8 the source asks whether

$$\mathbb{E}\mathcal{R}_n(\sigma) \asymp \sigma_{n+1}$$

also holds when $\sigma_j \asymp a^j$, $0 < a < 1$; its Corollary 9 gives only $a^n \lesssim \mathbb{E}\mathcal{R}_n(\sigma) \lesssim n^2 a^n$ [1].

Remark 8. The case $\sigma_n \asymp n^{-\alpha} \ln^{-\beta}(n+1)$ with $\alpha > 1/2$ can be extended to $\sigma_n \asymp n^{-\alpha} \varphi(n)$ for any slowly varying function φ . Also in this case, we get $\mathbb{E}[\mathcal{R}_n(\sigma)] \asymp \sigma_{n+1}$.

We also discuss sequences of exponential decay. We have seen that $\mathbb{E}[\mathcal{R}_n(\sigma)] \asymp \sigma_{n+1}$ holds for sequences with sufficiently fast polynomial decay. It remains open whether the same holds for sequences of exponential decay. With Theorem 4 we obtain the following.

Corollary 9. *Assume that $\sigma_n \asymp a^n$ for some $a \in (0, 1)$. Then*

$$a^n \lesssim \mathbb{E}[\mathcal{R}_n(\sigma)] \lesssim n^2 a^n.$$

Figure 1: The complete open-question passage and adjacent Corollary 9 on source PDF page 8.

2 Proof intuition

Use only the first $n + 1$ coordinates. The n Gaussian measurements restricted to these coordinates have a one-dimensional kernel, and this random null line is already contained in the full kernel. A unit direction of the line is uniform on the sphere, hence has the law $g/\|g\|_2$ for an $(n + 1)$ -dimensional standard Gaussian g .

For geometric semiaxes, reversing coordinates splits the resulting line radius into two scales. Its numerator $\|g\|_2$ is of order $\sqrt{n}$, whereas the denominator is a geometrically weighted Gaussian square sum with bounded mean. A constant-probability intersection of these two elementary events therefore forces a $\sqrt{n}$ loss over the optimal radius. No analysis of the rest of the infinite-dimensional kernel is needed.

3 Counterexample theorem

Theorem 1. *Let $0 < a < 1$, and let the nonincreasing positive semiaxes satisfy*

$$c_0 a^j \leq \sigma_j \leq C_0 a^j \quad (j \geq 1)$$

for constants $c_0, C_0 > 0$. Then, for every integer $n \geq 22$,

$$\mathbb{E}\mathcal{R}_n(\sigma) \geq \frac{c_0 \sqrt{1-a^2}}{4\sqrt{2}} \sqrt{n+1} a^{n+1}. \quad (1)$$

Consequently

$$\frac{\mathbb{E}\mathcal{R}_n(\sigma)}{\sigma_{n+1}} \geq \frac{c_0 \sqrt{1-a^2}}{4\sqrt{2} C_0} \sqrt{n+1} \rightarrow \infty.$$

In particular, the source's proposed comparison $\mathbb{E}\mathcal{R}_n(\sigma) \asymp \sigma_{n+1}$ is false for every exponentially decaying sequence in the stated class.

Proof. Let A_n be the restriction of G_n to its first $n + 1$ columns. Almost surely $\text{rank } A_n = n$. Choose a unit vector v spanning $\ker A_n$ and extend it by zero after coordinate $n + 1$. Then $G_n v = 0$. The radius of $\mathcal{E}_\sigma$ on the line $\mathbb{R}v$ is

$$L_n(\sigma) := \left(\sum_{j=1}^{n+1} \frac{v_j^2}{\sigma_j^2} \right)^{-1/2}, \quad \mathcal{R}_n(\sigma) \geq L_n(\sigma).$$

The law of A_n is invariant under right multiplication by every orthogonal matrix in dimension $n + 1$. Thus its null line is Haar distributed, and we may write, in distribution,

$$v = \frac{g}{\|g\|_2},$$

where $g_1, \dots, g_{n+1}$ are independent $N(0, 1)$ variables. Since $\sigma_j \geq c_0 a^j$,

$$\begin{aligned} L_n(\sigma) &\geq c_0 \frac{\|g\|_2}{\left(\sum_{j=1}^{n+1} a^{-2j} g_j^2 \right)^{1/2}} \\ &= c_0 a^{n+1} \frac{\|g\|_2}{Q_n^{1/2}}, \quad Q_n := \sum_{r=0}^n a^{2r} g_{n+1-r}^2. \end{aligned}$$

Now

$$\mathbb{E}Q_n = \sum_{r=0}^n a^{2r} \leq \frac{1}{1-a^2},$$

so Markov's inequality gives

$$\mathbb{P}\left\{Q_n \leq \frac{4}{1-a^2}\right\} \geq \frac{3}{4}. \quad (2)$$

Also $\|g\|_2^2$ is chi-square with $n+1$ degrees of freedom. The elementary chi-square lower-tail estimate

$$\mathbb{P}\{\|g\|_2^2 < (n+1)/2\} \leq e^{-(n+1)/16}$$

implies, when $n \geq 22$,

$$\mathbb{P}\{\|g\|_2^2 \geq (n+1)/2\} \geq \frac{3}{4}. \quad (3)$$

No independence between the events in (2) and (3) is required: their intersection has probability at least $1/2$. On that intersection,

$$L_n(\sigma) \geq c_0 a^{n+1} \sqrt{\frac{n+1}{2}} \sqrt{\frac{1-a^2}{4}} = c_0 a^{n+1} \sqrt{\frac{(n+1)(1-a^2)}{8}}.$$

Taking expectations and multiplying by the intersection probability proves (1). Finally, $\sigma_{n+1} \leq C_0 a^{n+1}$ gives the displayed diverging ratio. $\square$

4 Sharper witness asymptotics

For the exact sequence $\sigma_j = a^j$, the random line itself has a clean limiting scale. Let $h_0, h_1, \dots$ be iid standard Gaussians and set $Q_\infty = \sum_{r \geq 0} a^{2r} h_r^2$. Reordering coordinates in the formula above gives a coupling under which

$$\frac{L_n}{\sqrt{n} a^{n+1}} = \frac{(n^{-1} \sum_{r=0}^n h_r^2)^{1/2}}{(\sum_{r=0}^n a^{2r} h_r^2)^{1/2}} \longrightarrow Q_\infty^{-1/2} \quad \text{almost surely.}$$

The limit has a finite, strictly positive mean: indeed $Q_\infty \geq h_0^2 + a^2 h_1^2$, whose inverse square root is integrable in two Gaussian dimensions. Fatou's lemma and $\mathcal{R}_n \geq L_n$ therefore yield

$$\liminf_{n \rightarrow \infty} \frac{\mathbb{E}\mathcal{R}_n(a^j)}{\sqrt{n} a^{n+1}} \geq \mathbb{E}Q_\infty^{-1/2} > 0.$$

The explicit elementary estimate in Theorem 1, rather than this limit statement, is the primary counterexample.

5 Verification, limitations, and novelty

The proof is finite-dimensional until its final inclusion in $\ker G_n$. It uses only orthogonal invariance, Markov's inequality, and a standard chi-square tail bound. A Monte Carlo script included in the packet confirms that the witness ratio $L_n/(\sqrt{n} a^{n+1})$ stabilizes numerically; this is a sanity check, not part of the proof.

The theorem answers the source's yes/no comparison completely and negatively. It does *not* determine the exact order of $\mathbb{E}\mathcal{R}_n(\sigma)$ between the new lower bound $\sqrt{n} a^n$ and the source upper bound $n^2 a^n$.

A bounded novelty audit on 11 August 2026 searched the run’s registry, solution, attempt, and proof-gap indexes; the exact arXiv id, title, authors, and open-problem wording; arXiv queries combining random sections, ellipsoids, and exponential decay; and all 18 works then indexed by OpenAlex as citing DOI 10.1090/tran/8502. No later answer or occurrence of this random-line obstruction was found. Nearby arXiv:2109.14504 concerns primarily polynomial semiaxes of generalized ℓ_p -ellipsoids, and the 2022 survey arXiv:2209.07266 still records only a polynomial-loss exponential upper bound. Novelty is plausible but remains subject to specialist review.

Human-review recommendation. Verify the inclusion of the finite-coordinate null line in the full kernel, the uniform-sphere law, and the constants in the Markov/chi-square event intersection. If those three points pass, classify this as a full negative answer.

References

- [1] A. Hinrichs, D. Krieg, E. Novak, J. Prochno, and M. Ullrich, *Random sections of ellipsoids and the power of random information*, Trans. Amer. Math. Soc. **374** (2021), no. 12, 8691–8713; arXiv:1901.06639, doi:10.1090/tran/8502.